

REM tutorial

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Part I

Introduction to relational event models

1 Core REMs

Relational event sequence: realization of a **marked point process**.

$$E = \{e_i = (s_i, r_i, t_i), i = 1, \dots, n\}, \quad (1)$$

where:

- s_i : **sender** of the i -th relational event;
- r_i : **receiver** of the i -th relational event;
- t_i : **time** of the i -th relational event;
- n : **number of events**.
- i : **index of a generic event**.

Longitudinal network / multivariate counting process [Perry and Wolfe, 2013]:

$$\{N_{sr}(t)\}_{s \in V^S, r \in V^R, t \geq 0}, \quad (2)$$

where:

- $N_{sr}(t)$: number of interactions from s to r until time t .
- V^S : **set of senders** in the system;
- V^R : **set of receivers** in the system;

If senders/receivers belong to the same group, then $V^S = V^R = V$ (**vertex set**), with $p = |V|$.

Doob-Meyer decomposition theorem:

$$N_{sr}(t) = \Lambda_{sr}(t) + M_{sr}(t) = \int_0^t \lambda_{sr}(u) du + M_{sr}(t)$$

where:

- $\Lambda_{sr}(t)$: **cumulative intensity process** (structural part);
- $M_{sr}(t)$: **Martingale residual process** (noisy part);
- $\lambda_{sr}(t)$: **hazard** (also: **rate, intensity**): instantaneous rate at which (s, r) occurs at t .

Relational event model (REM) [Bianchi et al., 2024]:

$$\lambda_{sr}(t) = W_{sr}(t) \times \lambda_0(t) \times \exp\{f(x_{sr}(t), t)\} \quad (3)$$

where:

- $W_{sr}(t)$: **risk indicator** – it is equal to 1 if (s, r) is at risk at time t ;
- $\lambda_0(t)$: **baseline hazard** (including global risk determinants);
- $f(x_{sr}(t), t)$: **event log-rate contribution**. E.g. **Linear contribution** $f(x_{sr}(t), t) = \beta \cdot x_{sr}(t)$
 - $f(\cdot, t)$: **contribution function** (also, **effect**);
 - $x_{sr}(t)$: edge-specific **covariates**;
 - t : **calendar time**.

The dyads (s, r) satisfying $W_{sr}(t) = 1$ compose the **risk set** $\mathcal{R}_t$.

Gillespie algorithm for simulating REMs:

$$\begin{aligned}\Delta T_{sr}(t) &\sim \text{Exponential}(\lambda_{sr}(t)), \\ T &= \min_{(s,r) \in \mathcal{R}_t} \Delta T_{sr}(t) \sim \text{Exponential}\left(\sum_{(s,r) \in \mathcal{R}_t} \lambda_{sr}(t)\right), \\ D &\sim \text{Multinomial}(\{p_{sr}(t)\}_{(s,r) \in \mathcal{R}_t}), \\ p_{sr}(t) &= \frac{\lambda_{sr}(t)}{\sum_{(s,r) \in \mathcal{R}_t} \lambda_{sr}(t)}, \quad (s,r) \in \mathcal{R}_t.\end{aligned}$$

where:

- $\Delta T_{sr}(t)$ is the **waiting time** (random variable) until an event involving sender s and receiver r occurs, evaluated at time t .
- T is the waiting time (random variable) until the next event, evaluated at time t .
- D (random variable) identifies the dyad (s, r) involved in the event occurring at time $t+T$.

Assumption: $\lambda_{sr}(t)$ does not change between t and $t+T$.

Exogenous & endogenous drivers: measured by means of **covariates** $x_{sr}(t)$. Covariates can be seen as aggregations/summaries of information derived from **attributes** $z_{sr}(t)$ of nodes and edges.

- **Exogenous covariates:** do not depend on the network of (previous) events.

E.g. **Work experience dissimilarity**, obtained as the absolute value of the difference in **work experience** between the two nodes involved in the event.

$$x_{sr}(t) = |WE_s - WE_r|$$

- **Endogenous covariates:** depend on the network of (previous) events and allow us to measure **relational mechanisms**.

E.g. **Reciprocity**.



- **Binary covariates:** check for the presence (in the previous event history) of **at least one event satisfying a condition** of interest.

E.g. Is there **at least one reciprocal event** in the past?

$$x_{sr}(t) = \mathbf{1}(\{(r, s, t^*) \mid t^* < t\} \neq \emptyset)$$

- **Count covariates:** count the **number of events satisfying a condition** of interest.

E.g. **How many reciprocal events** occurred in the past?

$$x_{sr}(t) = |\{(r, s, t^*) \mid t^* < t\}|$$

- **Time-aware covariates:** defined as a function of the **internal time** of the mechanism of interest (i.e., the difference between the current time and the time of the event satisfying the condition of interest in the previous event history).

E.g. **How recent is the reciprocal event** in the past?

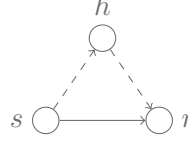
$$z_{sr}(t) = \begin{cases} t - \tau_{sr}, & \tau_{sr} = \text{time of the last event } (r, s), \\ \infty, & \text{otherwise,} \end{cases} \quad x_{sr}(t) = \exp(-z_{sr}(t))$$

Relational mechanisms can be **dyadic** or **triadic**, depending on the number of nodes involved in the structure of interest. Some examples:

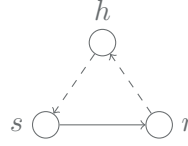
Repetition (dyadic):



Transitive closure (triadic):



Cyclic closure (triadic):



2 REM Inference

Assumption (this section only): **linear contribution function**, i.e., $f(x_{sr}(t), t) = \beta \cdot x_{sr}(t)$.

The counting process (18) is adapted to the **filtration** $\mathbb{W} = \{\mathcal{W}_t\}_{t \in \mathbb{R}^+}$. At time t , $\mathcal{W}_{t-}$ incorporates both exogenous information and information from events occurred at $t^* < t$.

Likelihood: constructed as joint probability of observing **data** (1) under **model** (19).

Preliminaries: Given a set of event times $t_1, t_2, \dots, t_n$, which are realizations of a point process:

- the associated counting process is a right-continuous function,
- that jumps by one at each event time and is constant otherwise,
- with conditional intensity function be $\lambda(t \mid \mathcal{W}_{t-})$

the joint probability density of observing exactly n event times in the interval $(0, t_n)$ is

$$f(t_1, t_2, \dots, t_n) = \prod_{i=1}^n \lambda(t_i \mid \mathcal{W}_{t_i-}) \exp \left\{ - \int_{t_{i-1}}^{t_i} \lambda(u \mid \mathcal{W}_{u-}) du \right\}.$$

2.1 Full likelihood approaches

Full likelihood [Butts, 2008]:

$$\mathcal{L}(\beta) = \prod_{i=1}^n p(t_i \mid \mathcal{W}_{t_i-}; \beta) \times p((s_i, r_i) \mid t_i, \mathcal{W}_{t_i-}; \beta) \quad (4)$$

$$= \prod_{i=1}^n \underbrace{\sum_{(s,r) \in \mathcal{R}_{t_i}} \lambda_{sr}(t_i \mid \mathcal{W}_{t_i-}; \beta) \exp \left[- \sum_{(s,r) \in \mathcal{R}_{t_i}} \int_{t_{i-1}}^{t_i} \lambda_{sr}(u \mid \mathcal{W}_u; \beta) du \right]}_{\text{Minimum Generalized Exponential Inter-arrival Time}} \underbrace{\frac{\lambda_{s_i r_i}(t_i \mid \mathcal{W}_{t_i-}; \beta)}{\sum_{(s,r) \in \mathcal{R}_{t_i}} \lambda_{sr}(t_i \mid \mathcal{W}_{t_i-}; \beta)}}_{\text{Multinomial Probability}} \quad (5)$$

Piecewise Constant Rates (PCR) assumption [Stadtfeld et al., 2017]: event rate is assumed to be constant within inter-arrival times $\Rightarrow$ Exponential **waiting time** $\Delta T_i = T_i - T_{i-1}$ with Poisson number of events in ΔT_i .

$$\mathcal{L}(\beta) \stackrel{\text{PCR}}{=} \prod_{i=1}^n \lambda_{s_i r_i}(t_i | \mathcal{W}_{t_i}^-; \beta) \prod_{(s,r) \in \mathcal{R}_{t_i}} \exp \left(- \lambda_{sr}(t_i | \mathcal{W}_{t_i}^-; \beta) (t_i - t_{i-1}) \right) \quad (6)$$

Letting $\Delta N_{s_i r_i} = N_{s_i r_i}(t_i) - N_{s_i r_i}(t_{i-1})$,

$$\mathcal{L}(\beta) \stackrel{\text{PCR}}{=} \prod_{i=1}^n \prod_{(s,r) \in \mathcal{R}_{t_i}} \left(\lambda_{sr}(t_i | \mathcal{W}_{t_i}^-; \beta) \right)^{\Delta N_{s_i r_i}} \exp \left(- \lambda_{sr}(t_i | \mathcal{W}_{t_i}^-; \beta) (t_i - t_{i-1}) \right) \quad (7)$$

is proportional to the likelihood of a **Poisson Regression**.

Full log-likelihood in practice (PCR)

Log-likelihood :

$$\ell(\beta) = \sum_{i=1}^n \sum_{(s,r) \in \mathcal{R}_{t_i}} \Delta N_{sr}(t_i) \log(\lambda_{sr}(t_i; \beta)) - \lambda_{sr}(t_i; \beta) (t_i - t_{i-1})$$

Inference technique : **Poisson regression;**

$$\Delta N_{sr}(t_i) | \mathbf{x}_{sr}(t_i) \stackrel{\text{iid}}{\sim} \text{Poisson}(\mu_{sr}(t_i)), \quad \log(\mu_{sr}(t_i)) = \beta \cdot \mathbf{x}_{sr}(t_i) + \log(t_i - t_{i-1})$$

Data overview & R code :

stratum	event	sender	receiver	time	dist	logtm
1	1	31419	31255	0.0218023	1.6832458	0.0218023
1	0	31261	31071	0.0218023	1.8376884	0.0218023
1	0	31937	32050	0.0218023	2.7475273	0.0218023
1	0	31937	31937	0.0218023	0.0000000	0.0218023
1	0	31071	31023	0.0218023	1.9742199	0.0218023
1	0	31937	31272	0.0218023	2.0297259	0.0218023
1	0	32217	31255	0.0218023	3.8173386	0.0218023
1	0	32604	31966	0.0218023	2.4291296	0.0218023
1	0	32050	31937	0.0218023	2.7357951	0.0218023
1	0	31285	31324	0.0218023	0.5883421	0.0218023

```
glm(event ~ log_dist + offset(logtm),
family = poisson(link = "log"), data = poisson_dat)
```

2.2 Partial likelihood approaches

Consider the second term in Equation (5): their joint product is also known as **partial likelihood** [Cox, 1975].

$$\mathcal{L}^P(\beta) = \prod_{i=1}^n \frac{\lambda_{s_i r_i}(t_i | \mathcal{W}_{t_i}^-; \beta)}{\sum_{(s,r) \in \mathcal{R}_{t_i}} \lambda_{sr}(t_i | \mathcal{W}_{t_i}^-; \beta)} = \prod_{i=1}^n \frac{\exp \{ \beta \cdot \mathbf{x}_{s_i r_i}(t_i) \}}{\sum_{(s,r) \in \mathcal{R}_{t_i}} \exp \{ \beta \cdot \mathbf{x}_{sr}(t_i) \}} \quad (8)$$

Partial log-likelihood in practice

Log-likelihood :

$$\ell^P(\beta) = \log \mathcal{L}^P(\beta) = \sum_{i=1}^n \left[\beta \cdot \mathbf{x}_{s_i r_i}(t_i) - \log \left(\sum_{(s,r) \in \mathcal{R}_{t_i}} \exp \{ \beta \cdot \mathbf{x}_{sr}(t_i) \} \right) \right],$$

Inference technique : **CoxPH regression;**

$$\hat{\beta} = \arg \max_{\beta} \ell^P(\beta)$$

Data overview & R code :

stratum	event	sender	receiver	time	dist
1	1	31419	31255	0.0218023	1.6832458
1	0	31261	31071	0.0218023	1.8376884
1	0	31937	32050	0.0218023	2.7475273
1	0	31937	31937	0.0218023	0.0000000
1	0	31071	31023	0.0218023	1.9742199
1	0	31937	31272	0.0218023	2.0297259
1	0	32217	31255	0.0218023	3.8173386
1	0	32604	31966	0.0218023	2.4291296
1	0	32050	31937	0.0218023	2.7357951
1	0	31285	31324	0.0218023	0.5883421

```
rem(event ~ dist, data = cox_dat, method = "clogit", stratum = "stratum")
```

2.3 Sampled partial likelihood approaches

Nested case-control dataset [Borgan et al., 1995, Vu et al., 2015, Lerner and Lomi, 2020]:

$$D_m = \{(e_i, \mathcal{SR}_{t_i}), \mathcal{SR}_{t_i} \subset \mathcal{R}_{t_i}, |\mathcal{SR}_{t_i}| = m + 1, i = 1, \dots, n\}, \quad (9)$$

where:

- $\mathcal{SR}_{t_i}$: **sampled risk set** at t_i , including (s_i, r_i) and m randomly sampled dyads in $\mathcal{R}_{t_i}$.
- m : **number of sampled non-events**.
- Event history and related sampling information are incorporated in a **filtration** $\mathbb{F} = \{\mathcal{F}_t\}_{t \in \mathbb{R}^+}$ where $\mathcal{F}_t = \mathcal{W}_t \cup \sigma(\mathcal{SR}_{t_i}; t_i \leq t)$.

Relational event model with sampling:

$$\lambda_{sr, \mathcal{SR}}(t) = \lambda_{sr}(t) \cdot \pi_t(\mathcal{SR} | (s, r)), \quad \pi_t(\mathcal{SR} | (s, r)) = \frac{m}{|\mathcal{R}_t| - 1} \cdot 1_{\{(s, r) \in \mathcal{SR}, \mathcal{SR} \subset \mathcal{R}_t\}} \quad (10)$$

Sampled partial likelihood:

$$\mathcal{L}^S(\beta) = \prod_{i=1}^n \frac{\lambda_{s_i r_i}(t_i; \mathcal{F}_{t_i}^-; \beta) \cdot \pi_{t_i}(\mathcal{SR}_{t_i} | (s_i, r_i))}{\sum_{(s, r) \in \mathcal{SR}_{t_i}} \lambda_{sr}(t_i; \mathcal{F}_{t_i}^-; \beta) \cdot \pi_{t_i}(\mathcal{SR}_{t_i} | (s, r))} = \prod_{i=1}^n \frac{\exp\{\beta \cdot \mathbf{x}_{s_i r_i}(t_i)\}}{\sum_{(s, r) \in \mathcal{SR}_{t_i}} \exp\{\beta \cdot \mathbf{x}_{sr}(t_i)\}}, \quad (11)$$

that coincides with the likelihood of a **conditional logistic regression** of a stratum of size $m + 1$ with the first $k = 1$ observations being the cases.

Sampled partial log-likelihood in practice

Log-likelihood :

$$\ell^S(\beta) = \log \mathcal{L}^S(\beta) = \sum_{i=1}^n \left[\beta \cdot \mathbf{x}_{s_i r_i}(t_i) - \log \left(\sum_{(s, r) \in \mathcal{SR}_{t_i}} \exp\{\beta \cdot \mathbf{x}_{sr}(t_i)\} \right) \right]$$

Inference technique : Conditional logistic regression;

$$\Delta N_{sr}(t_i) \mid \sum_{s' r' \in \mathcal{SR}_i} \Delta N_{s' r'}(t_i) = 1 \sim \text{Categorical}(\pi_{sr}(t_i))$$

$$\pi_{sr}(t_i) = \frac{\exp\{\beta \cdot \mathbf{x}_{sr}(t_i)\}}{\sum_{(s', r') \in \mathcal{SR}_{t_i}} \exp\{\beta \cdot \mathbf{x}_{s' r'}(t_i)\}}$$

Data overview & R code :

stratum	event	sender	receiver	time	dist
1	1	31023	31261	0.0102023	2.042388
1	0	31023	31419	0.0102023	2.405412
1	0	32050	31966	0.0102023	3.271848
1	0	31246	31324	0.0102023	1.066778
1	0	31023	33200	0.0102023	2.078691
1	0	31419	32217	0.0102023	3.833369
2	1	31419	31255	0.0140662	1.683246
2	0	31071	31324	0.0140662	2.088897
2	0	32050	33200	0.0140662	2.527966
2	0	31313	32217	0.0140662	3.753660

```
rem(event ~ dist, data = sampled_dat, method = "clogit", stratum = "stratum")
```

2.4 Case-1-control partial likelihood approaches

Consider (11) in the particular case of $m = 1$ [Boschi et al., 2025a, Filippi-Mazzola and Wit, 2024].

$$\mathcal{L}^S(\beta) \stackrel{m=1}{=} \prod_{i=1}^n \frac{\exp\{\beta \cdot \mathbf{x}_{s_i r_i}(t_i)\}}{\exp\{\beta \cdot \mathbf{x}_{s_i r_i}(t_i)\} + \exp\{\beta \cdot \mathbf{x}_{s_i^* r_i^*}(t_i)\}} = \prod_{i=1}^n \frac{\exp\{\beta \cdot \Delta \mathbf{x}_i\}}{1 + \exp\{\beta \cdot \Delta \mathbf{x}_i\}} = \prod_{i=1}^n \text{logistic}(\beta \cdot \Delta \mathbf{x}_i),$$

where:

- (s_i^*, r_i^*) : the **sampled dyad** at time t_i , drawn from the risk set $\mathcal{R}_{t_i}$ ($m = 1$);
- $\Delta \mathbf{x}_i = \mathbf{x}_{s_i r_i}(t_i) - \mathbf{x}_{s_i^* r_i^*}(t_i)$: the difference between covariates evaluated for the event and the sampled non-event at time t_i .

Equation (2.4) coincides with the likelihood of a **degenerate logistic regression**:

- without intercept term,
- with constant response equal to 1,
- with covariates equal to the difference between covariates evaluated for the event and the sampled non-event at time t_i .

Case-1-control likelihood in practice

Log-likelihood :

$$\ell^S(\beta) \stackrel{m=1}{=} \log \mathcal{L}^S(\beta) = \sum_{i=1}^n \left[\beta \cdot \mathbf{x}_{s_i r_i}(t_i) - \log \left(\sum_{(s,r) \in \{(s_i, r_i), (s_i^*, r_i^*)\}} \exp\{\beta \cdot \mathbf{x}_{sr}(t_i)\} \right) \right]$$

Inference technique : Degenerate logistic regression;

$$Y_i | \Delta \mathbf{x}_i \stackrel{\text{iid}}{\sim} \text{Bernoulli}(\pi_i), \quad \text{logit}(\pi_i) = \beta \cdot \Delta \mathbf{x}_i, \quad y_i = 1 \quad i = 1, \dots, n; \quad \text{no intercept} \quad (12)$$

Data overview & R code :

stratum	time	sender_ev	receiver_ev	sender_nv	receiver_nv	dist_ev	dist_nv
1	0.0015464	31218	31255	31419	31419	1.6821306	0.000000
2	0.0029034	31272	31324	33202	31966	1.7032925	2.873395
3	0.0096929	31937	32050	32050	31262	2.7475273	2.547647
4	0.0452324	31261	31262	31313	31218	1.1820339	2.201659
5	0.0517514	31272	31272	31419	31218	0.0000000	1.875181
6	0.0524939	33200	31261	31255	33202	0.5538851	1.243866
7	0.0531366	31285	31419	31419	33200	1.6052291	1.784903
8	0.0572758	32604	32217	31966	31262	3.0713035	2.821915
9	0.0628786	31285	33202	31071	31023	1.0922588	1.974220
10	0.0676353	31255	32050	31297	31272	2.4686079	1.655558

```
rem(~ dist, data = dat_gam, method = "gam")
```

3 Non-Linear REMs

Additive relational event model (example): consider Equation (19) with

$$f(x_{sr}(t), v_{sr}(t), t) = \underbrace{\beta \cdot x_{sr}^{(1)}(t)}_{\text{fixed linear}} + \underbrace{\beta(t) \cdot x_{sr}^{(2)}(t)}_{\text{time-varying}} + \underbrace{f^{(3)}(x_{sr}^{(3)}(t))}_{\text{non-linear}} + \underbrace{\gamma \cdot v_{sr}(t)}_{\text{random}} \quad (13)$$

as a contribution function, including four types of effects:

1. **Fixed linear effect** (FLE): β is time-homogenous and is not a function of $x_{sr}^{(1)}(t)$;
2. **Time-varying effect** (TVE): $\beta(t)$ is a non-linear function of time;
3. **Non-linear effect** (NLE): $f^{(3)}(x_{sr}^{(3)}(t))$ is a non-linear function of $x_{sr}^{(3)}(t)$;
4. **Random effect** (RE): $\gamma = (\gamma_s)_{s \in V}$, γ_s is a Gaussian random variable;

This model can be fitted using a degenerate logistic regression defined in (12).

Time-varying effect. In its simplest form, a **smooth function of time** can be represented as a linear combination of basis functions of **calendar time**.

$$\beta(t) \approx \sum_{j=1}^q \theta_j b_j(t), \quad (14)$$

where:

- q : **basis dimension**;
- $\{b_j\}_j$: set of **basis functions**;
- θ_j : **basis coefficients**;

TVE in practice

[Boschi et al., 2025a]

Contribution to the log-event rate :

$$f(x_{sr}(t), v_{sr}(t), t) = \dots + \underbrace{\sum_{j=1}^q \theta_j b_j(t) \times x_{sr}^{(2)}(t)}_{\text{TVE}} + \dots$$

Contribution to the case-1-control partial likelihood :

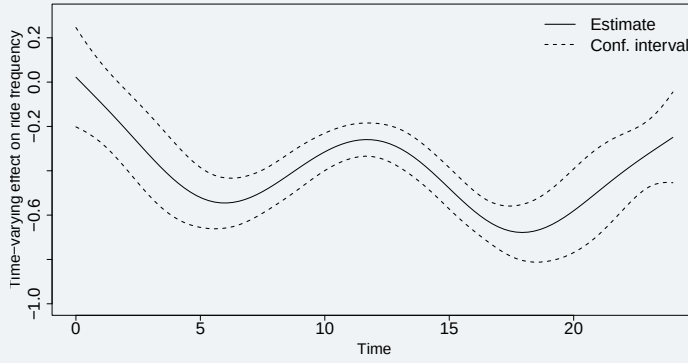
$$\mathcal{L}^S(\theta) \stackrel{m=1}{=} \prod_{i=1}^n \frac{e^{\dots + [\sum_{j=1}^q \theta_j b_j(t_i)] \Delta x_i^{(2)} + \dots}}{1 + e^{\dots + [\sum_{j=1}^q \theta_j b_j(t_i)] \Delta x_i^{(2)} + \dots}} \quad (+ \text{smoothing penalty})$$

Data overview & R code :

stratum	time	sender_ev	receiver_ev	sender_nv	receiver_nv	distPT_ev	distPT_nv
1	0.0014511	32217	32604	32604	31937	3.8203727	6.2539595
2	0.0015844	31937	31285	32217	31246	1.7116404	3.8203727
3	0.0081908	31324	31218	31272	31261	1.3001624	2.2255955
4	0.0117318	31313	31272	31297	32604	2.3462775	0.7931991
5	0.0145487	31285	32217	31261	31285	2.3303198	1.4511050
6	0.0183787	33200	31313	31262	31419	4.1927036	1.1147747
7	0.0199062	31218	31285	31966	31255	1.9189675	1.8293264
8	0.0203390	31023	31272	31023	31324	0.8268842	0.8268842
9	0.0211421	31966	31246	33202	31966	1.8293264	3.2714157
10	0.0224279	33200	32050	31297	31071	4.1927036	0.7931991

```
rem(~ tv(distPT), time = "time", data = dat_gam_tv, method = "gam")
```

Estimated effect (example):



Non-linear effect: In its simplest form, a **smooth function of a covariate** can be represented as a linear combination of basis functions of the **covariate** itself.

$$f(x_{sr}(t)) \approx \sum_{j=1}^q \theta_j b_j(x_{sr}(t)) \quad (15)$$

where (as before):

- q : **basis dimension**;
- $\{b_j\}_j$: set of **basis functions**;
- θ_j : **basis coefficients**;

NLE in practice

[Filippi-Mazzola and Wit, 2024]

Contribution to the log-event rate :

$$f(x_{sr}(t), v_{sr}(t), t) = \dots + \underbrace{\sum_{j=1}^q \theta_j b_j(x_{sr}^{(3)}(t))}_{\text{NLE}} + \dots$$

Contribution to the case-1-control partial likelihood :

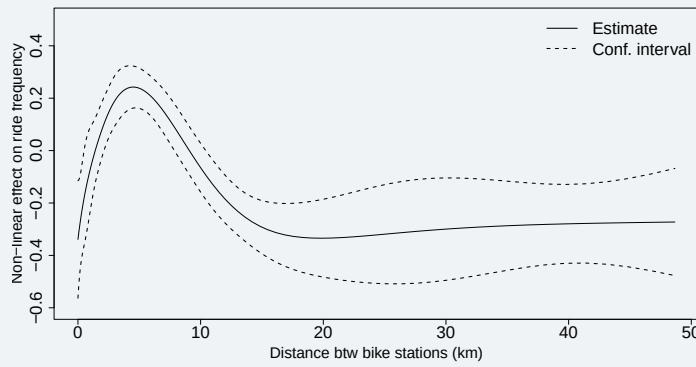
$$\mathcal{L}^S(\theta) \stackrel{m=1}{=} \prod_{i=1}^n \frac{e^{\dots + \left[\sum_{j=1}^q \theta_j [b_j(x_{s_i r_i}^{(3)}(t_i)) - b_j(x_{s_i^* r_i^*}^{(3)}(t_i))] \right]} + \dots}{1 + e^{\dots + \left[\sum_{j=1}^q \theta_j [b_j(x_{s_i r_i}^{(3)}(t_i)) - b_j(x_{s_i^* r_i^*}^{(3)}(t_i))] \right]} + \dots} \quad (+ \text{smoothing penalty})$$

Data overview & R code :

	stratum	time	sender_ev	receiver_ev	sender_nv	receiver_nv	dist_ev	dist_nv
1	1	0.0013772	33202	32604	32604	31937	2.712905	2.328253
2	2	0.0014538	31937	31285	32217	31246	1.687139	3.801985
3	3	0.0019790	33200	31937	31255	31261	1.833861	1.058137
4	4	0.0044957	31246	31937	31218	31937	1.466260	1.946053
5	5	0.0074578	31324	31218	31272	31261	1.603420	1.458615
6	6	0.0105801	31272	31324	31272	31937	1.703293	2.039791
7	7	0.0112535	31966	31262	31285	32217	2.821915	3.821223
8	8	0.0113760	33202	32217	31272	31261	3.849168	1.458615
9	9	0.0117538	31313	31262	33202	32604	2.015569	2.712905
10	10	0.0142903	31246	33200	31966	31255	1.096610	2.766130

```
rem(~ nl(dist), data = dat_gam_nl, method = "gam")
```

Estimated effect (example):



Random effect: can be efficiently estimated as **o-dimensional splines**, with basis functions taking the value 1 when the level of the random factor is present and 0 otherwise.

$$\sum_{s' \in V} \gamma_{s'} \mathbb{1}\{s' = s\} \quad \gamma = (\gamma_1, \dots, \gamma_{|V|})$$

RE in practice

[Boschi et al., 2025a]

Contribution to the log-event rate :

$$f(x_{sr}(t), v_{sr}(t), t) = \dots + \underbrace{\sum_{s' \in V} \gamma_{s'} \mathbb{1}\{s = s'\}}_{\text{RE}} + \dots$$

Contribution to the case-1-control partial log-likelihood :

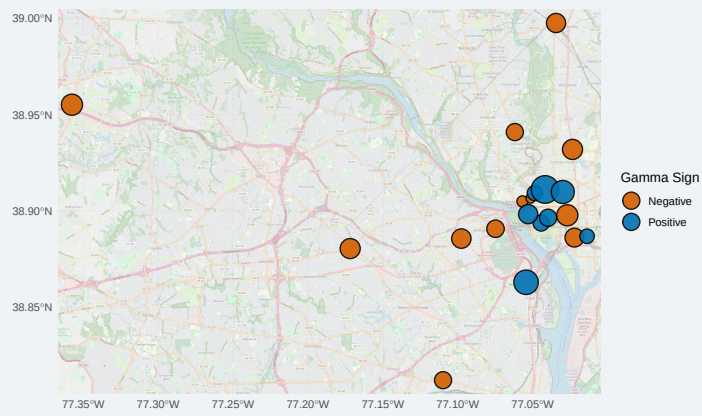
$$\ell^S(\gamma) = \log \mathcal{L}^S(\gamma) \stackrel{m=1}{=} \sum_{i=1}^n \log \left(\frac{e^{\dots + \gamma_{s_i} - \gamma_{s_i^*} + \dots}}{1 + e^{\dots + \gamma_{s_i} - \gamma_{s_i^*} + \dots}} \right) - \frac{1}{2\sigma_s^2} \gamma \cdot \gamma$$

Data overview & R code :

stratum	time	sender_ev	receiver_ev	sender_nv	receiver_nv
1	0.0013772	33202	32604	32604	31937
2	0.0014538	31937	31285	32217	31246
3	0.0019790	33200	31937	31255	31261
4	0.0044957	31246	31937	31218	31937
5	0.0074578	31324	31218	31272	31261
6	0.0105801	31272	31324	31272	31937
7	0.0112535	31966	31262	31285	32217
8	0.0113760	33202	32217	31272	31261
9	0.0117538	31313	31262	33202	32604
10	0.0142903	31246	33200	31966	31255

```
rem(~ re(sender), data = dat_gam_r, method = "gam")
```

Estimated effect (example):



Part II

All you need to know about relational hyperevent modeling

1 Core RHEMs

Relational hyper event sequence: realization of a **marked point process**.

- Sequence of **undirected hyperevents** [Lerner et al., 2025b]:

$$E = \{e_i = (S_i, t_i), i = 1, \dots, n\}, \quad (16)$$

where:

- S_i is a subset of the **vertex set** V , composed of the entities involved simultaneously in the i -th relational hyperevent, i.e. the **participants**;
- t_i : **time** of the i -th relational hyperevent;

E.g. **Les misérables**, whose chapters can be considered as hyperevents where actors co-appear.

$$E = \{(\{MY, NP, MB\}, t_1), (\{MY, ME, MB\}, t_2), (\{MY\}, t_3), (\{MY, ME, CL, GE, MC, MB\}, t_4), (\{MY, ME, MB\}, t_5), \dots\}$$

- Sequence of **directed hyperevents** [Lerner et al., 2025a, Boschi et al., 2025b]:

$$E = \{e_i = (S_i, R_i, t_i), i = 1, \dots, n\}, \quad (17)$$

- S_i is a subset of the **sender set** V^S , representing the group of senders of the i -th relational hyperevent;
- R_i is a subset of the **receiver set** V^R , representing the group of receivers of the i -th relational hyperevent;
- t_i : **time** of the i -th relational hyperevent;

E.g. **Scientific innovation**, where each paper publication is a hyperevent (a group of authors publishes a paper citing a set of papers).

$$E = \{(\{2, 3\}, \{A, B\}, t_1 = 1950), (\{1, 2, 4\}, \{A, C\}, t_2 = 1951), (\{4\}, \{A, C\}, t_3 = 1953)\}$$

Remark: an undirected hyperevent can be represented as a directed hyperevent with an empty receiver set, i.e. $(S_i, t_i) = (S_i, R_i = \emptyset, t_i)$ is also referred to as a **meeting**.

In both cases, n is the **number of hyperevents** and i the **index of a generic hyperevent**.

Longitudinal network / multivariate counting process [Perry and Wolfe, 2013]:

$$\{N_{SR}(t)\}_{S \subseteq V^S, R \subseteq V^R, t \geq 0}, \quad (18)$$

where:

- $N_{SR}(t)$: number of interactions from S to R until time t .
- V^S : **set of senders** in the system;
- V^R : **set of receivers** in the system;

If senders/receivers belong to the same group, then $V^S = V^R = V$ (**vertex set**), with $p = |V|$.

Doob-Meyer decomposition theorem:

$$N_{SR}(t) = \lambda_{SR}(t) + M_{SR}(t) = \int_0^t \lambda_{SR}(u) du + M_{SR}(t)$$

where:

- $\Lambda_{SR}(t)$: **cumulative intensity process** (structural part);
- $M_{SR}(t)$: **Martingale residual process** (noisy part);
- $\lambda_{SR}(t)$: **hazard** (also: **rate, intensity**): instantaneous rate at which (S, R) occurs at t .

Relational hyper event model (RHEM) [Lerner et al., 2021, Lerner and Lomi, 2023]:

$$\lambda_{SR}(t) = W_{SR}(t) \times \lambda_0(t) \times \exp \left\{ f(x_{SR}(t), t) \right\} \quad (19)$$

where:

- $W_{SR}(t)$: **risk indicator** – it is equal to 1 if (S, R) is at risk at time t ;
- $\lambda_0(t)$: **baseline hazard** (including global risk determinants);
- $f(x_{SR}(t), t)$: **event log-rate contribution**. E.g. **Linear contribution** $f(x_{SR}(t), t) = \beta \cdot x_{SR}(t)$
 - $f(\cdot, t)$: **contribution function** (also, **effect**);
 - $x_{SR}(t)$: edge-specific **covariates**;
 - t : **calendar time**.

(S, R) satisfying $W_{SR}(t) = 1$ compose the **risk set** $\mathcal{R}_t$.

1.1 Exogenous & endogenous drivers for undirected hyperevents

Exogenous and endogenous drivers are measured by means of **covariates** $x_S(t)$. Covariates can be seen as aggregations or summaries of information derived from **attributes** $z_s(t)$ of nodes in subsets $S' \subseteq S$.

- **Exogenous covariates**: do not depend on the network of (previous) events. One can consider both the **main effect** of an exogenous feature and the effect of related **homophily**.

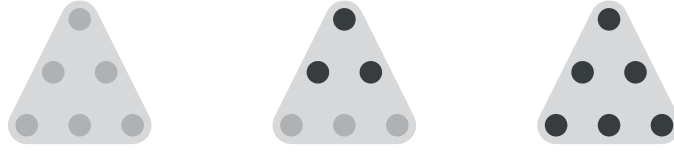
E.g. **Gender**. Gender is a binary¹ feature that can be evaluated at each node. The covariate capturing the main effect can be computed as the average of the feature over the nodes in the set S .

$$\text{avg.z}(S, t) = \sum_{s \in S} z_s(t) / |S|$$

Gender **dissimilarity** can be captured using the following covariate. A negative effect of this covariate indicates the presence of **homophily**.

$$\text{diff.z}(S, t) = \sum_{\{s, s'\} \in \binom{S}{2}} \frac{|z_s(t) - z_{s'}(t)|}{\binom{|S|}{2}}$$

¹only for mathematical purposes

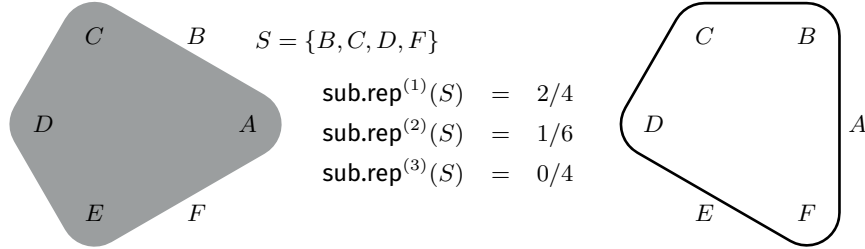


$$\begin{array}{lll} \text{avg.z}(S_1) & = & 0 \\ \text{diff.z}(S_1) & = & 0 \end{array} \quad \begin{array}{lll} \text{avg.z}(S_2) & = & 0.5 \\ \text{diff.z}(S_2) & = & \frac{9}{15} \end{array} \quad \begin{array}{lll} \text{avg.z}(S_3) & = & 1 \\ \text{diff.z}(S_3) & = & 0 \end{array}$$

- **Endogenous covariates:** depend on the network of (previous) events and allow us to measure **relational mechanisms**.

E.g. **Subset repetition:** the average number of prior joint events over all subsets $S' \subseteq S$ of size ρ . As before, we evaluate the attribute (in this case, the **formation of a hyperedge** in the past) for all possible subsets and then **aggregate** the results to obtain the covariate.

$$\begin{aligned} \text{hy.deg}(S', t) &= \sum_{e_i: t_i < t} \mathbf{1}(S' \subseteq S_i) \\ \text{sub.rep}^{(\rho)}(S, t) &= \sum_{S' \in \binom{S}{\rho}} \frac{\text{hy.deg}(S', t)}{\binom{|S|}{\rho}} \end{aligned}$$



2 Non-linear RHEMs

2.1 Directed hyperevents

This section reviews some examples of **directed hyperevents**, specifically referring to the paper publication hyperevent defined above. Data concerning directed hyperevents manifest in the form of Equation (17).

2.2 Exogenous & endogenous drivers for directed hyperevents

Endogenous covariates: depend on the network of (previous) events and allow us to measure **relational mechanisms**.

E.g. **Subset repetition.** The definition can be extended in the presence of directed hyperevents by considering the average number of prior joint events involving a subset of senders and a subset of receivers of a given size in the past event history.

$$\begin{aligned} \text{cite}^{\text{aut-pap}}(S, R, t) &= \sum_{t_i < t} \omega(t - t_i) \cdot \mathbf{1}_{\{S \subseteq S_i \cap R \subseteq R_i\}} \\ \text{subrep}^{(\rho, \ell)}(S, R, t) &= \sum_{(S', R') \in \binom{S}{\rho} \times \binom{R}{\ell}} \frac{\text{cite}^{\text{aut-pap}}(S', R', t)}{\binom{|S|}{\rho} \times \binom{|R|}{\ell}} \end{aligned} \quad (20)$$

We propose defining subset repetition as an aggregation of the **author–paper citation attribute** (see Equation (??) for more information). However, subset repetition is a very general concept and can be extended well beyond the notion of citation.

2.3 Beyond Linearity

What can break Linearity and Time-Homogeneity?

Saturation : individuals face cognitive overload and become unable to effectively process additional information.

Forgetfulness : propensity to lose track of past details over time.

External Changes : shifts in the environment, technology, or social norms that alter how factors influence the system.

We adapt the notation introduced in Equations (14) and (15) to distinguish the indices of the basis functions when combining the two types of functions.

1. **Fixed linear effect:**

$$f^{\text{FLE}}(x_{SR}(t), t) = \beta \cdot x_{SR}(t) \quad (21)$$

2. **Time-varying effect:**

$$f^{\text{TVE}}(x_{SR}(t), t) = \alpha(t) \cdot x_{SR}(t), \quad \alpha(t) = \sum_{l=1}^L \alpha_l a_l(t) \quad (22)$$

3. **Non-linear effect:**

$$f^{\text{NLE}}(x_{SR}(t)) = \sum_{q=1}^Q \beta_q b_q(x_{SR}(t)) \quad (23)$$

Joint time-varying non-linear effect: coefficients of basis evaluations of covariates are not fixed but vary over time.

$$f^{\text{TVNLE}}(x_{SR}(t), t) = \sum_{q=1}^Q \left(\sum_{l=1}^L \alpha_{ql} a_l(t) \right) b_q(x_{SR}(t)) \quad (24)$$

To improve the **model fitting stability** when using these types of effects it is possible to:

1. Apply a **monotonic transformation** to each covariate x :

$$\dot{x} = 1 - \exp\left(-\frac{x}{\epsilon}\right)$$

- Choose ϵ so that the transformed covariate (evaluated for events) is as close as possible to a uniform distribution.
- Apply the same transformation to both events and non-events.

2. Transform event times t using the **empirical cumulative distribution function**:

$$\dot{t} = \text{ecdf}_t(t)$$

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